YEAR 7 · UNIT 1 · HOMEWORK 4

Squares, Cubes and Roots

Mathematics 1.6 — Square Roots and Cube Roots · 1.5 — Tests for Divisibility

Name: ____________________________ **Class:** ____________________________

Date given: ______________________ **Date due:** ________________________

How to do this homework

- Write your answers **on this sheet**. **No calculator.**
- Always **write the calculation, not just the answer**.
- Do **A1** first. You need those two lists for the whole packet.
- In **Section B** you use last week's divisibility tests to find roots.

Section A · Mathematics 1.6 — Squares and Cubes

Remember from the lesson

5^2 means $5 \times 5 = 25$. We say **"five squared"**. The small 2 counts **how many fives**, so 5^2 is **not** 5×2. 5^3 means $5 \times 5 \times 5 = 125$. We say **"five cubed"**.

A **root** goes **backwards**. $5^2 = 25$, so the **square root** of 25 is 5. $5^3 = 125$, so the **cube root** of 125 is 5.

A1 The two lists

(a) Fill in the **twelve square numbers**. Use your notebook.

n	1	2	3	4	5	6	7	8	9	10	11	12
n^2												

(b) Now the **cube numbers**.

n	1	2	3	4	5	6	10
n^3							

(c) One number is on **both** lists. Which one? ______________

A2 Square it, cube it, root it

(a) Work out **both**. Then write which one is bigger.

7^2 = ________ 7×2 = ________ Bigger: ________

(b) Use your lists from A1.

(i) 11^2 = ______	(iv) 10^3 = ______	(vii) $\sqrt[3]{27}$ = ______
(ii) 12^2 = ______	(v) $\sqrt{81}$ = ______	(viii) $\sqrt[3]{216}$ = ______
(iii) 4^3 = ______	(vi) $\sqrt{144}$ = ______	(ix) $\sqrt[3]{64} - \sqrt{100}$ = ______

(c) One answer in (b) is **negative**. Which one? ____________

A3 Say it, then write it

Word help — these words tell you what to do

square 9	$9^2 = 9 \times 9$
the square of 9	the same thing: $9^2 = 81$
the square root of 81	9, because $9 \times 9 = 81$
cube 4	$4^3 = 4 \times 4 \times 4$
the cube root of 64	4, because $4 \times 4 \times 4 = 64$
a square number	a number on your first list
consecutive	one after another, nothing missed out. 6 and 7 are consecutive. 36 and 49 are consecutive **square** numbers.

Write the calculation for each sentence, then work it out. **Calculation first.**

1. Square 11.
   Calculation: ______________________ Answer: ________

2. Find the square root of 121.
   Calculation: ______________________ Answer: ________

3. Subtract the cube root of 27 from the square of 8.
   Calculation: ______________________ Answer: ________

A4 Trap the root

When the root is not a whole number

45 is not a square number, so $\sqrt{45}$ is not whole. But 45 is between **36** and **49**, and those two **are** on the list. So $\sqrt{45}$ is between **6** and **7** — two **consecutive** whole numbers.

Write the two consecutive whole numbers each root is between.

(a) _____ < $\sqrt{30}$ < _____ (c) _____ < $\sqrt{110}$ < _____

(b) _____ < $\sqrt{70}$ < _____ (d) _____ < $\sqrt[3]{50}$ < _____

(e) Two **consecutive square numbers** add up to **85**. Which two? ______________________

Their square roots are _______ and _______

Section B · Maths 1.5 and 1.6 — Using the Tests on a Root

A square is a pair. A cube is three copies.

36 = 6 × 6. Both numbers in the pair are the **same**, so every factor inside one is inside the other as well. If 3 divides a square number, then 9 divides it too.

A cube is three copies: 8 = 2 × 2 × 2. So a factor is in there **three times**. If 2 divides a cube number, then 8 divides it too.

Rule 1 divides by 3 but **not** by 9 ⇒ **not** a square number

Rule 2 divides by 2 but **not** by 4 ⇒ **not** a square number

Rule 3 divides by 2 but **not** by 8 ⇒ **not** a cube number

Careful. These rules can only say NO. If a number passes, that does **not** prove it is a square.

B1 Can it be a square number?

Use the tests and fill in every box. In the last column write **NO** if a rule says no, or **maybe** if it passes. If the number is **odd**, put a dash in the **by 2** and **by 4** columns.

Number	Digit sum	by 3?	by 9?	by 2?	by 4?	Square?
132						
441						
162						
1296						
108						

(g) One number passed **every** rule. Which one? ______________

Check it against your A1 list. It is between ______ and ______ — two square numbers.

So is it a square number? ___________ (yes or no)

B2 Split it, then find the root

Split the number into two pieces you know

Split the number into two pieces. Find the root of each piece. Then **multiply the two roots**. **Good pieces:** 9, 4 and 100 are squares (roots 3, 2, 10); 8 and 1000 are cubes (roots 2 and 10).

Example. $\sqrt{576}$. The digits add to $5 + 7 + 6 = 18$, so **9 divides it**. $576 \div 9 = 64$. Now $\sqrt{9} = 3$ and $\sqrt{64} = 8$, so $\sqrt{576} = 3 \times 8 =$ **24**. Check: $24 \times 24 = 576$.

Now do these. The number to divide by is given. Fill in the rest.

Find	Divide by	Split it	The two roots	Answer
$\sqrt{441}$	9			
$\sqrt{1296}$	9			
$\sqrt[3]{1728}$	8			
$\sqrt[3]{8000}$	1000			

B3 The challenge: 5832

5832, from Homework 3

In Homework 3 you ticked **5832** for 2, 3, 4, 6, 8 and 9. So it passes **every** rule in B1 — but it is **not** a square number. It **is** a cube number. Find its cube root.

Step 1. 5832 divides by 8. Do the division. **Tip: dividing by 8 is the same as halving three times.** $5832 \div 8 =$ ____________

Step 2. Your answer to Step 1 is $9 \times 9 \times 9$, so its cube root is __________

Step 3. $\sqrt[3]{8} =$ _______, so $\sqrt[3]{5832} =$ _______ $\times$ _______ $=$ _________

Step 4. Check your answer. $18 \times 18 =$ __________ then _______ $\times 18 =$ ___________

B4 Word problem

The photograph. Mr Bowen puts **400 students** in a **perfect square**. The photographer climbs onto the roof. How many students are in each row? Which test did you use to split 400?